

DILPREET SINGH

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SUMMARY

Electrical and Computer Engineering student at TIET with hands-on experience building full-stack web applications, relational database systems, and IoT data pipelines. Comfortable across the stack — backend API design, database modelling, and frontend dashboard development. Writes clean, self-authored code and can defend every architectural decision.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala, Punjab, India

B.E, Electrical and Computer Engineering

May 2027

Cambridge Global School, Patiala, Punjab, India

Senior Secondary

May 2023

SKILLS

Languages: Python, C++, SQL

Frameworks & Tools: Flask, SQLAlchemy, Jinja2, SQLite, Git, REST APIs, Chart.js

Concepts: Database Design, Relational Modelling, CRUD Operations, API Development, MVC Architecture, DBMS, OOP

Familiar With: HTML, CSS, Embedded Systems (ESP32, Arduino), MS Excel

PROJECTS

PlacementLog: Full-Stack Placement Tracking System

- Architected and built a full-stack web application using Flask and SQLite with SQLAlchemy ORM, managing relational data across 3 core modules — companies, applications, and interview question repositories
- Designed a normalised relational schema with foreign key constraints to maintain referential integrity across interconnected entities and prevent data redundancy
- Developed RESTful routes handling full CRUD operations for all modules, with Jinja2-templated frontend and shared form components for consistent UI across views
- Built an analytics dashboard using Chart.js visualising 5 key metrics including application pipeline progress, company status distribution, and round-level data

Syllabus Progress Tracker: Academic Management System

- Developed a Flask-based multi-user system with relational database workflows supporting syllabus tracking across multiple user roles and academic modules
- Implemented data integrity constraints and reconciliation logic to eliminate duplicate records; restructured information flow to reduce manual overhead

Air Quality Monitoring System: ESP32 IoT Device

- Designed a real-time sensor data pipeline on ESP32 processing continuous readings from MQ135 and MQ2 sensors across 2 air quality parameters
- Implemented non-blocking threshold logic triggering automated breach alerts — demonstrating interrupt-driven programming and real-time systems design
- Applied systematic sensor calibration and root cause analysis to reduce false positives and improve measurement reliability across monitored parameters

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Head of Marketing

Aug 2025 – Jul 2026

SSA VIRSA, TIET

- Directed marketing operations for a 200-member cultural society; coordinated cross-functional teams across 4+ events and managed end-to-end stakeholder communication

Mentor

Aug 2024 – Jul 2025

FROSH (Admission Cell), TIET

- Mentored incoming students within an admission cell serving approximately 2,000 freshers; structured onboarding guidance across diverse student cohorts

Executive Committee Member

Aug 2024 – Jul 2025

SATURNALIA, TIET

- Supported planning and execution for one of TIET's flagship technical festivals; coordinated across teams under tight delivery timelines

Executive Committee Member

Aug 2024 – Jul 2025

URJA, TIET